



DRILLHOLE RECORD

CONTRACT NO. GE/2022/08

HOLE NO.

DH 5

SHEET

1

of

3

PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk

METHOD **ROTARY**

CO-ORDINATES

E 829036.19

N 827484.26

TASK ORDER NO.

GE/2022/08.11B

MACHINE

SD44

DATE

09.12.2022

to

14.12.2022

FLUSHING MEDIUM

WATER

ORIENTATION **VERTICAL**

GROUND LEVEL

+171.34 mPD

Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description
09.12.2022	SW													
1			80	76					1	+171.34	0.00			Pale brown (10YR 6/3), clayey sandy SILT with much angular to subangular fine to coarse gravel of rock fragments and with some rootlets. (FILL)
									T2-101	+170.84	0.50			Dark grey (10YR 4/1) mottled light brownish grey and yellowish brown, angular to subangular, COBBLE and much coarse gravel of rock fragments. (FILL)
2	SW 1.55m		80	67										
	PW								T2-101		1.55			
3			80	0						+169.04	2.30			Light grey (10YR 7/1) and light yellowish brown (10YR 6/4), angular to subangular, clayey silty sandy fine to coarse GRAVEL and much cobble of rock fragments. (COLLUVIUM)
4			80	95					2	+167.94	3.40			Yellow (10YR 8/8) dappled grey, angular to subangular, slightly clayey silty sandy fine to coarse GRAVEL of rock fragments. (COLLUVIUM)
5									3		3.40			
									4		4.40			
									5		4.50			
									6		4.60			
6			80	100					7		5.40			
	PW 5.90m								8	+165.44	5.90			
	HW		80	97					T2-101		6.65			Grey (10YR 6/1) mottled light brownish grey and yellowish brown, angular to subangular, medium to coarse GRAVEL and much cobble of rock fragments. (COLLUVIUM)
09.12.2022		0.42 at 1800								+164.58	6.75			
10.12.2022		1.72 at 0800												
7			80	93	36	22	14.3		T2-101		7.03		III	Moderately weak to moderately strong, grey and yellowish brown, moderately decomposed, coarse ash TUFF. Joints are very closely to closely spaced, rough planar and rough stepped, iron and manganese oxide stained, dipping at 0° to 10°, 10° to 20° and 65° to 75°.
8										+163.26	8.08			
			80	100	83	72			T2-101		9.11		III	From 7.50m to 8.08m: Moderately weak. Moderately strong to strong, grey dappled yellowish brown and light olive green, moderately decomposed, coarse ash TUFF. Joints are closely to medium spaced, occasionally very closely spaced, rough planar and rough stepped, iron and manganese oxide stained, chlorite coated, dipping at 0° to 10°, 15° to 25°, 45° to 65° and 65° to 75°.
9														
10			80	100	100	100	3.6		T2-101	+161.34	10.00			From 11.40m to 11.60m, 15.60m to 15.73m and

- SMALL DISTURBED SAMPLE
- LARGE DISTURBED SAMPLE
- U76 SAMPLE
- PISTON SAMPLE (76mm)
- MAZIER SAMPLE
- SPT LINER SAMPLE
- WATER SAMPLE
- U100 SAMPLE

- STANDARD PENETRATION TEST
- IN-SITU VANE SHEAR TEST
- PACKER TEST
- PERMEABILITY TEST
- PRESSUREMETER TEST
- BOREHOLE TELEVIEWER
- PIEZOMETER TIP
- STANDPIPE TIP

LOGGED

C. Chan

DATE

20.12.2022

CHECKED

R. Chu

DATE

21.12.2022

REMARKS

- An inspection pit was excavated to 0.50m deep by hand tools.
- A standpipe was installed to 6.50m.
- A piezometer was installed with tip at 21.00m.



DRILLHOLE RECORD

CONTRACT NO. GE/2022/08

HOLE NO. **DH 5**SHEET **2** of **3**

PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk

METHOD		ROTARY		CO-ORDINATES		TASK ORDER NO.								
MACHINE		SD44		E 829036.19 N 827484.26		GE/2022/08.11B								
FLUSHING MEDIUM		WATER		ORIENTATION		VERTICAL								
GROUND LEVEL						+171.34 mPD								
Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description
11			80	85	57	53	NR		T2-101	10.63			III	16.98m to 17.20m: No recovery, assumed to be completely decomposed TUFF.
12			80	100	90	72	>20		T2-101	+159.94 +159.74	11.40 11.60		V	From 12.15m to 12.32m: Tuffaceous SILTSTONE.
13			80	100	90	72	5.4		T2-101	12.05	12.20		III	From 14.05m to 14.15m, 14.95m to 15.12m and 15.43m to 15.60m: Weak and highly decomposed. (Angular, slightly sandy fine to coarse GRAVEL of highly decomposed tuff fragments)
14			80	100	90	79	NI		T2-101	13.51			IV	From 14.15m to 14.80m, 15.73m to 16.41m and 16.62m to 16.98m: Strong and slightly decomposed.
15			80	94	68	64	6.3		T2-101	+157.29 +157.18	14.05 14.15		II	
16			80	66	21	0	NI		T2-101	+156.54 +156.39 +156.22	14.80 14.95 15.12		III	
17			80	100	89	81	2.9		T2-101	+155.91 +155.74 +155.61	15.35 15.60 15.73		IV	
18			80	100	89	81	8.8		T2-101	+154.93 +154.72	16.41 16.62		III	
19			80	42	42	42	NR		T2-101	+154.36 +154.14	16.98 17.20		V	
20			80	0					T2-101	17.20			IV	Weak, reddish yellow (5YR 6/6), highly decomposed, coarse ash TUFF. (Angular, slightly silty clayey sandy fine to coarse GRAVEL of highly decomposed tuff fragments)
			80	70	0	0	NI		T2-101	+153.04 +152.71 +152.57	18.30 18.63 18.77		IV	Weak to moderately weak, yellowish brown (10YR 5/4) mottled grey, highly decomposed, coarse ash TUFF. (Angular, clayey silty sandy fine to coarse GRAVEL and some cobble of highly decomposed tuff fragments)
			80	100	0	0	NI		T2-101	+152.17	19.17		V	
			80	84	15	15	>20		T2-101	19.17	19.36		III	From 18.63m to 18.77m: No recovery, assumed to be completely decomposed TUFF.
			80	84	15	15	>20		T2-101	20.00			III	Moderately weak to moderately strong, light

SMALL DISTURBED SAMPLE

LARGE DISTURBED SAMPLE

U78 SAMPLE

PISTON SAMPLE (76mm)

MAZIER SAMPLE

SPT LINER SAMPLE

WATER SAMPLE

U100 SAMPLE

STANDARD PENETRATION TEST

IN-SITU VANE SHEAR TEST

PACKER TEST

PERMEABILITY TEST

PRESSUREMETER TEST

BOREHOLE TELEVIEWER

PIEZOMETER TIP

STANDPIPE TIP

LOGGED **C. Chan**
DATE **20.12.2022**
CHECKED **R. Chu**
DATE **21.12.2022**

REMARKS



DRILLHOLE RECORD

CONTRACT NO. GE/2022/08

HOLE NO.

DH 5

SHEET

3 of 3

PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk

METHOD ROTARY

CO-ORDINATES

TASK ORDER NO.

GE/2022/08.11B

MACHINE SD44

E 829036.19

N 827484.26

DATE

09.12.2022 to 14.12.2022

FLUSHING MEDIUM WATER

ORIENTATION VERTICAL

GROUND LEVEL

+171.34 mPD

Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description
21			80	75			NR		T2-101 16-20.20 11-20.80 20.90	+151.30 +151.14	20.04 20.20		III IV	brownish grey, moderately decomposed, coarse ash TUFF. Joints are very closely to closely spaced, rough planar and rough stepped, iron and manganese oxide stained, clayey silt coated, dipping at 0° to 10°, 10° to 20° and 65° to 75°.
			80	100	92	0	>20		T2-101 21.28	+150.44	20.90		III	From 20.04m to 20.20m: No recovery, assumed to be completely decomposed TUFF.
13.12.2022 14.12.2022		0.93 at 1800 8.60 at 0800	80	100	23	0	NI		T2-101 21.84	+149.93	21.41		IV	Weak, pale brown (10YR 6/3), highly decomposed, coarse ash TUFF. (Angular, slightly silty sandy fine to coarse GRAVEL and some cobble of highly decomposed tuff fragments)
			80	84	22	13	NR		T2-101	+149.50	21.84		III	
			80				13.3		T2-101 22.73	+149.05	22.29		V	Moderately weak to moderately strong, light olive grey stripped yellowish brown, moderately decomposed, SILTSTONE.
23		HW 22.73m	80	100	100	100			T2-101 24.25	+148.91	22.43		III	Joints are very closely to closely spaced, rough planar and rough stepped, iron and manganese oxide stained, dipping at 0° to 10°, 10° to 20°, 35° to 45° and 65° to 75°.
24			80	100	100	100			T2-101 25.73	+148.61	22.73		II	From 21.41m to 21.84m: Weak to moderately weak and highly decomposed. (Angular, medium to coarse GRAVEL of highly decomposed siltstone fragments)
25			80	100	100	100			T2-101 27.24	+148.05	22.43		V	From 22.29m to 22.43m: No recovery, assumed to be completely decomposed SILTSTONE.
26			80	100	100	100			T2-101 27.99	+148.05	22.43		V	Strong to very strong, dark grey and bluish grey dappled pink and light grey, slightly decomposed, coarse ash TUFF. Joints are very widely spaced, rough planar, calcite and chlorite coated, dipping at 5° to 15°.
27			80	100	100	100			T2-101	+148.05	22.43		V	From 22.43m to 22.73m: Moderately strong and moderately decomposed.
28			80	100	100	100			T2-101	+148.05	22.43		V	From 22.43m to 22.73m and 24.96m to 25.55m: Tuffaceous SILTSTONE.
29			80	100	100	100			T2-101	+148.05	22.43		V	From 24.64m to 24.74m: With some quartz veins (<10mm thick).
30			80	100	100	100			T2-101	+143.35	27.99		V	End of hole at 27.99 m.

- SMALL DISTURBED SAMPLE
- LARGE DISTURBED SAMPLE
- U76 SAMPLE
- PISTON SAMPLE (76mm)
- MAZIER SAMPLE
- SPT LINER SAMPLE
- WATER SAMPLE
- U100 SAMPLE

- STANDARD PENETRATION TEST
- IN-SITU VANE SHEAR TEST
- PACKER TEST
- PERMEABILITY TEST
- PRESSUREMETER TEST
- BOREHOLE TELEVIEWER
- PIEZOMETER TIP
- STANDPIPE TIP

LOGGED

C. Chan

DATE

20.12.2022

CHECKED

R. Chu

DATE

21.12.2022

REMARKS